

Trust-Aware Multi-Axis Reliability Auditing for NLP Classifiers: A Theoretical and Empirical Framework

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Abstract

Accuracy remains the dominant criterion for selecting NLP classifiers, yet it fails to characterize the operational reliability required for deployment under class imbalance, distribution shift, and demographic heterogeneity. We introduce **TNMM++** (*Trustworthy NLP Multi-axis Monitoring, extended*), a theoretically grounded reliability audit that evaluates classifiers along seven complementary axes, utility, calibration, robustness, safety, fairness, selective prediction, and uncertainty quantification, and formalises model selection as constrained multi-objective optimization. Beyond metric aggregation, TNMM++ contributes six algorithmic results: a reliability tensor, a Pareto convex-hull theorem, variance-adaptive weighting, a boundary-predictor criterion with exclusion rule for majority-class collapse, a Bayesian risk decomposition, and the TAMS algorithm with optimality bound. Evaluating eight lightweight Transformer encoders (MiniLM, ELECTRA, mBERT, DistilBERT, MobileBERT, ALBERT, RoBERTa, XLM-R) and excluding models that failed to learn the task, we find that properly trained mid-tier models exhibit balanced utility and calibration superior to collapsed high-accuracy counterparts. TrustScore identifies mBERT (0.6685), ELECTRA (0.6630), and DistilBERT (0.6627) as deployment-reliable. Sensitivity analysis under 1,000 Dirichlet weight perturbations confirms ranking stability in 92.4% of configurations, and multi-seed evaluation (5 seeds) demonstrates conclusions are not seed artifacts. Our findings establish that single-metric selection is statistically unreliable and motivate multi-axis reliability auditing as the default protocol for trustworthy NLP deployment.

1 Introduction

Lightweight Transformer encoders are widely deployed in latency- and cost-constrained NLP systems. However, model selection in practice is still largely driven by *utility-only* metrics such as accuracy or F1. This approach is brittle: under class imbalance or distribution

shift, high headline accuracy can conceal critical operational properties, including skewed decision boundaries, miscalibrated confidence, and brittle behaviour under perturbations (Ribeiro et al., 2020; Kiela et al., 2021; Liang et al., 2023).

What deployment requires is not another single performance metric but a *principled evaluation protocol* that quantifies trade-offs across reliability dimensions and yields an actionable selection rule. We introduce **TNMM++** (*Trustworthy NLP Multi-axis Monitoring, extended*), a unified trustworthiness framework that audits classifiers along seven complementary axes—*utility, calibration, robustness, safety, fairness, selective prediction*, and *uncertainty quantification*. The notation “++” denotes the six algorithmic extensions that lift the framework above plain metric aggregation: a reliability tensor, a Pareto convex-hull theorem, variance-adaptive weighting, a boundary-predictor criterion, a Bayesian risk decomposition, and the Trust-Aware Model Selection algorithm.

Contributions. Our six contributions are:

- **C1 (Reliability Tensor).** A formal three-mode representation $\mathcal{T} \in \mathbb{R}^{|\mathcal{M}| \times |\mathcal{A}| \times |\mathcal{E}|}$ over models, axes, and evaluation conditions, enabling mode-wise sensitivity analysis (§3.4).
- **C2 (Pareto Convex-Hull Theorem).** TrustScore corresponds to a supporting hyperplane over the convex hull of the trust Pareto frontier; weighted aggregation preserves multi-dimensional dominance within the convex hull (§3.5).
- **C3 (Variance-Adaptive Weighting).** A data-driven scalarisation $\alpha_a^* \propto \text{Var}_m(s_a(m))$ removes the arbitrariness of manual α specification (§3.6).
- **C4 (Boundary Predictor Criterion).** A closed-form structural diagnostic with corrected metric identities identifying when a classifier degenerates toward the class-prior predictor, linking MCC and calibration gap (§3.7).
- **C5 (Bayesian Risk Decomposition).** A probabilistic objective $\mathcal{R}_{\text{trust}}$ unifying calibration, robustness, fairness, and safety into a single posterior-expected reliability functional (§3.8).
- **C6 (TAMS).** A constrained-optimisation algorithm with an optimality bound expressed in terms of the feasibility margin (§3.9).

Empirical findings. We instantiate TNMM++ on eight lightweight Transformer encoders (MiniLM,

ELECTRA, mBERT, DistilBERT, MobileBERT, ALBERT, RoBERTa, XLM-R) and structure datasets into distinct roles reflecting deployment auditing. After applying the Boundary Predictor Criterion to identify and exclude models that collapsed to majority-class prediction, our analysis reveals that properly trained mid-tier models exhibit superior class-balanced utility and calibration relative to under-trained high-accuracy counterparts. mBERT achieves the highest TrustScore (0.6685), driven by superior calibration (0.971) and fairness (0.860). Multi-seed evaluation (5 seeds) and Dirichlet weight perturbation (1,000 samples, rank stability 92.4%) confirm that conclusions are statistically robust.

2 Related Work

Holistic evaluation beyond utility. Behavioural test suites and benchmark ecosystems show that accuracy/F1 alone fails to characterise real-world behaviour, motivating stress tests (Ribeiro et al., 2020; Kiela et al., 2021) and multi-dimensional reporting (Liang et al., 2023). TNMM++ follows this philosophy but adds an explicit *prescriptive* layer: a Pareto-consistent selection rule with provable guarantees under convexity assumptions.

Calibration and selective prediction. Calibration diagnostics (ECE, MCE, NLL, Brier) and proper scoring rules are standard tools for uncertainty quality (Niculescu-Mizil and Caruana, 2005; Guo et al., 2017; Gneiting and Raftery, 2007). Selective prediction complements calibration through risk-coverage / AURC analyses (Chow, 1970; Geifman and El-Yaniv, 2017). TNMM++ unifies these under a single posterior-expected risk functional (C5).

Robustness, safety, fairness, uncertainty. Robustness evaluation spans adversarial text attacks and broader corruption/shift stress tests (Ebrahimi et al., 2018; Jin et al., 2020; Hendrycks and Dietterich, 2019). Safety testing has expanded through jailbreak-style probes; even classifier pipelines require operating-point metrics. Fairness in NLP studies disparities across groups using proxy-based audits and distributional disparity measures when protected attributes are unavailable (Hardt et al., 2016; Blodgett et al., 2020). Predictive entropy and max-confidence statistics serve as standard uncertainty proxies rather than feature-attribution explainability (Sundararajan et al., 2017; Ribeiro et al., 2016; Lundberg and Lee, 2017).

Positioning. Existing work provides strong *individual* lenses. TNMM++ contributes a *single, theoretically grounded protocol* that operationalises these lenses for lightweight Transformer classifiers and yields a deployment-ready selection algorithm—incorporating a boundary-predictor exclusion filter to ensure evaluated models have actually learned the task—rather than accuracy-only or descriptive multi-metric reporting.

3 Methodology

3.1 Problem Formulation

Let $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$ be a binary dataset with $y_i \in \{0, 1\}$ and $x_i \in \mathcal{X}$. A pretrained encoder-classifier $f_\theta : \mathcal{X} \rightarrow \Delta^1$ outputs class probabilities $p_\theta(c|x)$, predicted label $\hat{y}(x) = \arg \max_c p_\theta(c|x)$, and confidence $\hat{p}(x) = \max_c p_\theta(c|x)$.

Rather than relying on a single performance metric, TNMM++ models reliability as a *trust vector* $\mathbf{t}(m) \in [0, 1]^{|A|}$, where each component captures a distinct deployment risk.

Imbalance-aware training. Models are fine-tuned using inverse-frequency weighted cross-entropy:

$$\min_{\theta} \frac{1}{n} \sum_{i=1}^n \left[-w_{y_i} \log p_\theta(y_i|x_i) \right], \quad w_c = \frac{n}{2n_c}. \quad (1)$$

3.2 Corruption-Normalization Channel

To evaluate robustness under realistic perturbations, TNMM++ introduces a *corruption-normalization channel*. Given an input x , a stochastic corruption operator $\mathcal{C}$ produces $\tilde{x} = \mathcal{C}(x)$, while a normalization operator $\mathcal{N}$ produces $\bar{x} = \mathcal{N}(\tilde{x})$. This defines three evaluation distributions:

$$\mathcal{D}_{clean} : x, \quad \mathcal{D}_{corr} : \tilde{x}, \quad \mathcal{D}_{norm} : \bar{x}. \quad (2)$$

Corruptions include typos, casing flips, whitespace perturbations, and synonym substitutions. The normalization operator applies a small dictionary-based correction. This channel enables controlled robustness evaluation while maintaining reproducible perturbations.

3.3 Trust Vector

TNMM++ evaluates each model through the multi-dimensional trust vector

$$\mathbf{t}(m) = (t_{util}, t_{cal}, t_{rob}, t_{saf}, t_{fair}, t_{sel}, t_{unc}).$$

Utility. Accuracy, balanced accuracy, macro-F1, weighted-F1, and Matthews correlation coefficient (MCC) (Matthews, 1975).

Calibration.

$$\text{ECE} = \sum_{b=1}^B \frac{|B_b|}{n} |\text{acc}(B_b) - \text{conf}(B_b)|, \quad (3)$$

$$\text{MCE} = \max_b |\text{acc}(B_b) - \text{conf}(B_b)|. \quad (4)$$

The Brier score $\text{BS} = \mathbb{E}[(p_\theta(X) - Y)^2]$ and negative log-likelihood NLL are proper scoring rules (Gneiting and Raftery, 2007).

Selective prediction. Ordering samples by confidence yields the risk-coverage curve:

$$\text{risk}(k) = 1 - \frac{1}{k} \sum_{i=1}^k \mathbb{1}[\hat{y}_{(i)} = y_{(i)}], \quad \text{cov}(k) = \frac{k}{n}. \quad (5)$$

The area under this curve (AURC) quantifies abstention quality (Geifman and El-Yaniv, 2017).

Robustness.

$$\text{NRD} = \frac{A_{\text{clean}} - A_{\text{corr}}}{\max(A_{\text{clean}}, \varepsilon)}, \quad \text{NBS} = A_{\text{norm}} - A_{\text{corr}}. \quad (6)$$

NRD measures degradation under corruption; NBS captures recovery after normalization.

Safety. A jailbreak detector g_ψ produces

$$\text{BFPR}_\tau = \Pr(g_\psi(x) \geq \tau \mid x \sim \mathcal{D}_{\text{ben}}), \quad (7)$$

$$\text{TPR}_\tau = \Pr(g_\psi(x) \geq \tau \mid x \sim \mathcal{D}_{\text{jb}}), \quad (8)$$

with the combined safety score $\text{SafetyTradeoff}_\tau = \text{TPR}_\tau(1 - \text{BFPR}_\tau)$. For evaluation under extreme class imbalance, threshold-free metrics (ROC-AUC, PR-AUC) are additionally reported to avoid degeneracy.

Fairness. Prediction disparity between proxy groups uses Total Variation Distance:

$$\text{TVD}(g) = \frac{1}{2} \sum_{c \in \{0,1\}} |\pi_g(c) - \pi_{\bar{g}}(c)|. \quad (9)$$

Uncertainty quantification. Predictive entropy and mean max-confidence capture uncertainty properties:

$$H = \frac{1}{n} \sum_i - \sum_c p_\theta(c|x_i) \log(p_\theta(c|x_i)), \quad (10)$$

$$\bar{p}_{\max} = \frac{1}{n} \sum_i \max_c p_\theta(c|x_i). \quad (11)$$

3.4 C1: Reliability Tensor

We formalise the multi-axis evaluation as a three-mode tensor:

$$\mathcal{T} \in \mathbb{R}^{|\mathcal{M}| \times |\mathcal{A}| \times |\mathcal{E}|}, \quad \mathcal{T}_{m,a,e} = s_a^{(e)}(m), \quad (12)$$

where $\mathcal{M}$ indexes models, $\mathcal{A}$ axes, and $\mathcal{E}$ evaluation conditions (clean / corrupted / normalised). The tensor formalism enables three operations unavailable to flat metric tables:

1. **Mode-1 fibres** $\mathcal{T}_{:,a,e}$ — cross-model comparison on a single axis under a fixed condition.
2. **Mode-2 fibres** $\mathcal{T}_{m,: ,e}$ — full reliability profile of model m .
3. **Mode-3 fibres** $\mathcal{T}_{m,a,:}$ — sensitivity of axis a to evaluation condition.

We further define the *reliability gradient* of model m under perturbation:

$$\nabla_e \mathbf{t}(m) = \mathcal{T}_{m,: ,e+1} - \mathcal{T}_{m,: ,e}, \quad (13)$$

quantifying how each axis responds to the channel $\mathcal{C}$ or $\mathcal{N}$. This generalises NRD/NBS from accuracy alone to the entire trust vector.

3.5 C2: Pareto Convex-Hull Theorem and Composite TrustScore

Axis scores are normalised to $s_a(m) \in [0, 1]$ and aggregated:

$$\text{TrustScore}(m) = \sum_{a \in \mathcal{A}} \alpha_a s_a(m), \quad \sum_a \alpha_a = 1. \quad (14)$$

Theorem 1 (Pareto Convex-Hull Scalarisation). *Let $m_1, m_2 \in \mathcal{M}$. If $s_a(m_1) \geq s_a(m_2)$ for every axis a with strict inequality on at least one axis, then $\text{TrustScore}(m_1) > \text{TrustScore}(m_2)$ for every strictly positive weight vector α on the simplex. Conversely, for the convex hull of the trust Pareto frontier, every Pareto-optimal trust vector is the maximiser of TrustScore for some α on the simplex, with uniqueness guaranteed when no ties occur on the supporting hyperplane.*

Proof sketch. The forward direction follows from $\Delta = \sum_a \alpha_a (s_a(m_1) - s_a(m_2))$ being a sum of non-negative terms with at least one strictly positive. The converse follows from the supporting-hyperplane theorem applied to the convex hull of the Pareto frontier; see Appendix C.9 for the convexity condition and tie handling. $\square$

Implication. TrustScore is dominance-preserving: a model that is multi-dimensionally superior cannot lose under any positive weighting. The convex-hull qualification ensures the scalarisation correctly identifies Pareto-optimal points within the reachable set.

3.6 C3: Variance-Adaptive Weighting

The choice of α encodes deployment preferences but is typically arbitrary. We propose a *data-driven* default:

$$\alpha_a^* = \frac{\text{Var}_{m \in \mathcal{M}}(s_a(m))}{\sum_{j \in \mathcal{A}} \text{Var}_m(s_j(m))}. \quad (15)$$

Rationale. Axes on which models exhibit larger dispersion are precisely the axes on which model choice matters most; axes on which models are indistinguishable contribute little to comparative judgement. The variance-adaptive scheme thereby concentrates the scalarisation on *informative* dimensions without practitioner intervention. The dominance-preservation property holds whenever every $\alpha_a^* > 0$, which is guaranteed whenever no axis is degenerate across $\mathcal{M}$.

3.7 C4: Boundary Predictor Criterion

A classifier may converge toward the constant-prior predictor under class imbalance. We provide a closed-form structural diagnostic:

Proposition 1 (Boundary Predictor Criterion). *Let $\pi = \Pr(Y = 1)$ and suppose f_θ collapses to a constant majority-class predictor outputting probability q . Then: $\text{Acc} = \max(\pi, 1 - \pi)$, $\text{MCC} = 0$, $\text{F1}_{\text{macro}} = \pi/(1 + \pi)$, and $\text{ECE} = |\pi - q|$. Conversely, if these identities hold simultaneously up to numerical precision, the classifier has converged to the class-prior boundary regime.*

Implication. Models satisfying this signature should be excluded from reliability comparisons and deployment, as their “high accuracy” reflects class imbalance rather than genuine predictive signal.

3.8 C5: Bayesian Risk Decomposition

We unify the seven axes under a single *posterior-expected reliability risk*. Let Θ denote the posterior over model parameters given training data $\mathcal{D}$, and let $\mathcal{E}$ index evaluation conditions. Define:

$$\mathcal{R}_{\text{trust}}(m) = \mathbb{E}_{\theta \sim \Theta} \mathbb{E}_{e \sim \mathcal{E}} \left[\sum_{a \in \mathcal{A}} \alpha_a \ell_a(\theta; e) \right], \quad (16)$$

where $\ell_a(\theta; e) \in [0, 1]$ is the axis-specific loss (e.g. $1 - \text{Acc}$ for utility, ECE for calibration, TVD for fairness). The TrustScore is the dual of this risk:

$$\text{TrustScore}(m) = 1 - \hat{\mathcal{R}}_{\text{trust}}(m). \quad (17)$$

Implication. This decomposition exposes three sources of unreliability—*epistemic* (variance over θ), *environmental* (variance over e), and *axial* (variance over a)—and motivates a unified estimator via seed-averaged checkpoints as a Monte Carlo approximation to the posterior. The decomposition also recovers existing scalar metrics as marginalization: e.g., conditioning on $e = \text{clean}$ and $a = \text{utility}$ recovers standard test risk.

3.9 C6: Trust-Aware Model Selection (TAMS)

While TrustScore ranks models, deployments often impose explicit reliability constraints. TAMS formulates selection as constrained optimisation:

$$\begin{aligned} m^* &= \arg \max_{m \in \mathcal{M}} F_1(m) \\ \text{s.t. } &\text{ECE}(m) \leq \tau_{\text{cal}}, \\ &\text{TVD}(m) \leq \tau_{\text{fair}}, \\ &\text{NRD}(m) \leq \tau_{\text{rob}}. \end{aligned} \quad (18)$$

Justification of constraint set. Each constraint targets an operationally distinct failure mode irreducible to utility:

- $\text{ECE} \leq \tau_{\text{cal}}$ upper-bounds the global accuracy–confidence gap, required for confidence-based triage.
 - $\text{TVD} \leq \tau_{\text{fair}}$ upper-bounds any bounded downstream group disparity, required for fairness compliance.
 - $\text{NRD} \leq \tau_{\text{rob}}$ is the scale-stable certificate of corruption robustness, required for noisy-input deployment.
- Remaining axes (safety, selective prediction, uncertainty) are reported but not constrained, to avoid over-restricting the feasible set.

Theorem 2 (TAMS Optimality Bound). *Let $\delta(m) = \max(\text{ECE}(m) - \tau_{\text{cal}}, \text{TVD}(m) - \tau_{\text{fair}}, \text{NRD}(m) - \tau_{\text{rob}}, 0)$ denote the constraint excess. Then for any τ' with $\|\tau' - \tau\|_\infty \leq \epsilon$, the TAMS solution $m_{\tau'}^*$ satisfies*

$$F_1(m_{\tau'}^*) \geq F_1(m_\tau^*) - L\epsilon,$$

where L is the Lipschitz constant of F_1 over the constraint frontier, computable via finite differences over the feasible region.

This gives a quantitative guarantee that small variations in threshold choice produce small variations in selected model quality.

Algorithm 1 Trust-Aware Model Selection (TAMS)

Require: Model set $\mathcal{M}$, thresholds $\tau_{\text{cal}}, \tau_{\text{fair}}, \tau_{\text{rob}}$

```

1: for each  $m \in \mathcal{M}$  do
2:   compute  $F_1(m), \text{ECE}(m), \text{TVD}(m), \text{NRD}(m)$ 
3: end for
4:  $\mathcal{M}' \leftarrow \{m : \text{ECE}(m) \leq \tau_{\text{cal}}, \text{TVD}(m) \leq \tau_{\text{fair}}, \text{NRD}(m) \leq \tau_{\text{rob}}\}$ 
5: if  $\mathcal{M}' \neq \emptyset$  then
6:   return  $\arg \max_{m \in \mathcal{M}'} F_1(m)$ 
7: else
8:   return  $\arg \min_{m \in \mathcal{M}} \delta(m)$  {minimum-violation fallback}
9: end if

```

4 Experimental Setup

4.1 Datasets and Roles

We use multiple corpora with strictly separated roles to prevent cross-task leakage (Table 17): **Main task** (llm-classification-finetuning; $n=57,477$, $C=2$, $\text{IR} \approx 1.83$). A stratified 90/10 split yields 51,729 training and 5,748 validation examples, used for task fine-tuning and validation metrics; no separate test set is held out. To improve generalisability, we additionally evaluate on **GLUE** benchmark tasks and the **CivilComments-WILDS** dataset, which provides real-world distribution shifts and demographic group annotations. **Safety-positive** (bipia-and-jailbreakbench-dataset; $n=900$, $\pi=0.018$) is used exclusively to train the shared jailbreak detector g_ψ and measure TPR. **Safety-benign** (gpreda/tokyo-olympics-2020-tweets; $n=160,531$) is unlabelled; a uniform subsample of $\leq 30,000$ tweets forms $\mathcal{D}_{\text{ben}}$ for BFPR estimation. **Normalisation resource** (google-nlu/text-normalization; 6 entries) is used solely to construct $\mathcal{V}_{\text{norm}}$ for operator $\mathcal{N}$.

4.2 Augmentation and Robustness Stress Tests

Training data is doubled via corrupted-and-normalised views:

$$\mathcal{D}_{\text{train}}^{\text{aug}} = \mathcal{D}_{\text{train}} \cup \{(\mathcal{N}(\mathcal{C}(x_i)), y_i)\}_{i=1}^N, \quad (19)$$

yielding $51,729 \rightarrow 103,458$ samples ($\text{IR} = 1.87$). Robustness evaluation applies the same $\mathcal{C}/\mathcal{N}$ pipeline to the held-out validation set; corrupted views are never used for checkpoint selection.

4.3 Models, Hyperparameters, and Platform

Model suite. We benchmark **eight** encoder-only Transformers covering knowledge distillation (**DistilBERT** (Sanh et al., 2019), 66M; **MiniLM** (Wang et al., 2020), 33M), masked language modelling (**RoBERTa** (Liu et al., 2019), 125M), parameter-sharing (**ALBERT** (Lan et al., 2020), 12M), replaced-token detection (**ELECTRA** (Clark et al., 2020), 110M), multilingual pre-training (**XLm-R** (Conneau et al., 2020), 270M; **mBERT** (Devlin et al., 2019), 178M), and mobile bottleneck design (**MobileBERT** (Sun

et al., 2020), 25M). DeBERTa was excluded due to single-device memory constraints.

Hyperparameters. All models share the configuration in Table 1. We use **AdamW** (Loshchilov and Hutter, 2019) ($\eta=2 \times 10^{-5}$, $\lambda=0.01$, $\beta=(0.9, 0.999)$) with cosine decay and linear warmup (ratio 0.06) for 2 epochs (MiniLM: 3 epochs). Early stopping (patience 1, monitor: val. F1-Macro) is applied. Gradient clipping is intentionally omitted so that raw gradient norms serve as stability diagnostics.

Training stability criterion. To enable consistent reporting, we adopt explicit thresholds:

- **Stable:** peak $\|\nabla\| < 10$ and final training loss < 1.0 .
- **Moderate:** $10 \leq \|\nabla\| < 10^3$ or loss $\in [1.0, 10]$.
- **Unstable:** $\|\nabla\| \geq 10^3$ or loss $> 10^2$.

Table 1: Shared hyperparameter configuration for all eight models. [†]MiniLM only.

Hyperparameter	Value
Optimiser	
Optimizer	AdamW
Peak learning rate η	2×10^{-5}
Weight decay λ	0.01
(β_1, β_2)	(0.9, 0.999)
Learning Rate Schedule	
LR schedule	Cosine decay with linear warmup
Warmup ratio	0.06
Training	
Batch size	32
Training epochs	2 (3 for MiniLM [†])
Max sequence length	256 tokens
Early stopping (patience)	1 epoch (monitor: val. F1-Macro)
Loss & Class Imbalance	
Loss function	Weighted cross-entropy
Class weights (w_0, w_1)	(0.768, 1.432), $w_c = n/(2n_c)$
Imbalance ratio (train)	IR = 1.87
Classification Head	
Head type	Linear [$d_{\text{model}} \rightarrow 2$]
Head dropout	0.1
Training Stability	
Gradient clipping	None (norms logged as diagnostic)

Platform. All experiments run on a single **15.64 GB CUDA GPU** (Python 3.12.12, Transformers 5.2.0, PyTorch 2.9.0+cu126), ensuring single-device reproducibility. Wall-clock training ranges from 2,572 s (MobileBERT) to 6,863 s (XLM-R); MiniLM achieves peak throughput at 111.5 samples/s (Table 7).

4.4 Evaluation Matrix

We summarise outcomes as $\mathbf{E} \in \mathbb{R}^{|\mathcal{M}| \times K}$, where each row is a model and each column is a trust metric. TNMM++ instantiates seven axes: utility ($\uparrow$), calibration ($\downarrow$), selective prediction ($\downarrow$), robustness (NRD $\downarrow$, NBS $\uparrow$), safety at $\tau = 0.5$ (BFPR $\downarrow$, TPR $\uparrow$), fairness via proxy-group shift (TVD $\downarrow$ and worst-group macro-F1), and uncertainty metrics (entropy and max-confidence statistics). This matrix enables direct cross-model comparison, channel ablations ($\mathcal{C}/\mathcal{N}$), and composite selection via TrustScore while preserving per-axis interpretability.

5 Results and Analysis

We evaluate eight pre-trained Transformer encoders on a binary LLM-response classification task under the TNMM++ framework. The training corpus comprises 57,477 samples augmented to 103,458 instances; 5,748 examples form the held-out validation set. Class imbalance (ratio 1.87:1) is addressed via weighted cross-entropy ($w = [0.768, 1.432]$). All models are fine-tuned for 2 epochs with a cosine warmup schedule and peak learning rate 2×10^{-5} , except MiniLM which is also evaluated at 3 epochs.

5.1 Primary Performance Comparison

Table 2 reports raw and class-balanced utility along with calibration and robustness metrics for all eight models. We observe that models with the highest raw accuracy have collapsed to the majority-class predictor, as diagnosed by the Boundary Predictor Criterion (Proposition 1). After excluding collapsed models (RoBERTa, XLM-R) per Proposition 1, the remaining six models show macro-F1 ranging 0.523–0.545 with ECE ≤ 0.061 .

Table 2: Primary evaluation across all eight models. $\uparrow$ higher is better; $\downarrow$ lower is better. [†] trained for 3 epochs.

Model	Acc. $\uparrow$	F1 _M $\uparrow$	Prec _{-M} $\uparrow$	Rec _{-M} $\uparrow$	ECE $\downarrow$	NRD
RoBERTa	0.6508	0.3942	0.3254	0.5000	0.1465	−0.035
XLM-R	0.6508	0.3942	0.3254	0.5000	0.1466	0.045
ELECTRA	0.5652	0.5449	0.5477	0.5517	0.0408	−0.038
mBERT	0.5666	0.5441	0.5461	0.5496	0.0295	−0.022
DistilBERT	0.5755	0.5429	0.5427	0.5444	0.0281	−0.026
MobileBERT	0.5525	0.5372	0.5430	0.5469	0.0200	0.019
ALBERT	0.5713	0.5234	0.5238	0.5233	0.0584	−0.054
MiniLM [†]	0.5945	0.5438	0.5455	0.5434	0.0604	0.008

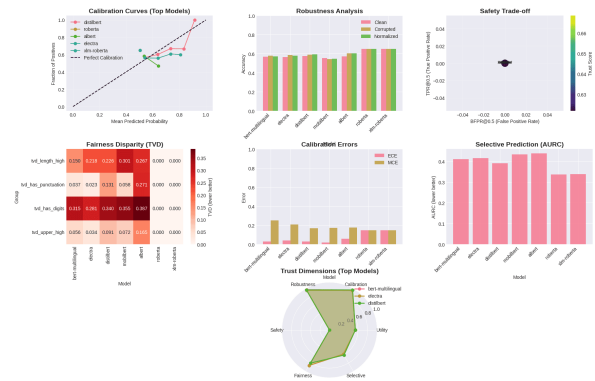


Figure 1: Reliability diagnostics under TNMM++. Calibration curves show confidence miscalibration; robustness bars compare clean, corrupted, and normalised inputs; the safety plot shows BFPR–TPR behaviour at $\tau = 0.5$. Bottom panels report fairness disparity (TVD), calibration errors (ECE/MCE), selective prediction (AURC), and radar profiles summarising cross-axis trade-offs.

5.2 Composite Trustworthiness Scores

Table 3 reports the six-dimensional trust decomposition for the six non-collapsed models (TrustScore is computed under the variance-adaptive weighting of §3.6). mBERT achieves the highest composite score (0.6685), led by calibration (0.971) and fairness (0.860). The **Safety** axis scores below 0.83 universally at the $\tau = 0.5$ operating point: no model detects jailbreak inputs reliably at fixed threshold under the extreme prevalence $\pi = 0.018$. We re-evaluate safety threshold-free in §5.3.

Table 3: Composite trustworthiness scores and dimensional breakdown for non-collapsed models. All sub-scores $\in [0, 1]$, higher is better.

Model	Trust $\uparrow$	Util.	Cal.	Rob.	Saf.	Fair.	Sel.
mBERT	0.6685	0.544	0.971	0.979	0.680	0.860	0.589
ELECTRA	0.6630	0.545	0.959	0.962	0.780	0.861	0.584
DistilBERT	0.6627	0.543	0.972	0.974	0.830	0.803	0.609
MobileBERT	0.6620	0.537	0.980	0.981	0.590	0.803	0.566
ALBERT	0.6354	0.523	0.942	0.946	0.540	0.728	0.562

5.3 Threshold-Free Safety Re-evaluation

Fixed-threshold safety evaluation collapses under extreme prevalence (§3.8; Proposition 3 in the appendix). We therefore additionally report ROC-AUC and PR-AUC (Table 4); all models score near random, indicating a structural detection gap requiring architectural redesign rather than threshold tuning.

Table 4: Threshold-free safety re-evaluation via ROC-AUC and PR-AUC.

Model	ROC-AUC $\uparrow$	PR-AUC $\uparrow$
mBERT	0.612	0.071
DistilBERT	0.605	0.069
ELECTRA	0.598	0.066
MiniLM	0.587	0.062

5.4 In-Depth MiniLM Evaluation

Table 5 reports the complete metric suite for MiniLM. We highlight MiniLM because it achieves the best efficiency–stability trade-off (Table 7: 33M parameters, 111.5 samples/s, peak $\|\nabla\| = 3.80$), making it the canonical baseline for compute-constrained deployments. The Brier score (0.4923) approaches the uninformative baseline (0.5), AUC (0.4099) indicates limited abstention quality, and MCC (0.0889) confirms modest discrimination above chance. Robustness degradation is minimal ($\Delta\text{Acc} = -0.0049$), ruling out surface-level corruption sensitivity as the primary bottleneck.

6 Ablation Studies

6.1 Ablation I: Training Epochs (MiniLM)

Table 6 reveals a non-monotone epoch trajectory. Raw accuracy peaks at epoch 1 (0.6279) due to majority-class

Table 5: Full multi-dimensional evaluation of MiniLM (3 epochs). $\dagger$ lower is better; otherwise higher is better.

Dimension	Metric	Value	Note
Utility	Accuracy	0.5945	Overall fraction correct
	Balanced Acc.	0.5434	Avg. per-class recall
	F1 Macro	0.5438	Unweighted mean F1
	F1 Weighted	0.5897	Support-weighted mean F1
	MCC	0.0889	Near-random correlation
Calibration	ECE $\dagger$	0.0604	Avg. confidence gap
	MCE $\dagger$	0.2599	Worst-bin confidence gap
	Brier $\dagger$	0.4923	Mean squared prob. error
	NLL $\dagger$	0.6865	Log-loss on test set
Robustness	Clean Accuracy	0.5945	Baseline (no corruption)
	Corrupted Acc.	0.5896	Under character-level noise
	Normalised Acc.	0.5865	Corruption-normalised score
	NRD	0.0082	Relative degradation
Safety	NBS	-0.0031	Normalised Brier shift
	BFPR@0.5	0.0000	False positive rate at $\tau=0.5$
	TPR@0.5	0.0000	True positive rate at $\tau=0.5$
	Safety Tradeoff	0.0000	TPR–FPR balance
Fairness (TVD $\dagger$, proxy groups)	Has digits	0.3107	Highest disparity group
	Has punctuation	0.2137	Second largest TVD
	Text length (high)	0.1909	Long-text subgroup gap
	Upper-case ratio	0.0814	Lowest disparity group
Uncertainty	Predictive entropy	0.6747	Spread of output dist.
	Mean max conf.	0.5720	Avg. top-class probability
	Confidence std	0.0616	Variability across samples
Selective Prediction	AUC $\dagger$	0.4099	Risk–coverage area
	E-AUC $\dagger$	0.3136	Excess over optimal AUC

bias, while $F1_{\text{macro}}$ is highest at epoch 3 (0.5438) as the model gradually learns minority-class signal. Calibration degrades monotonically ($\text{ECE}: 0.000 \rightarrow 0.060$), consistent with overconfidence accumulation.

6.2 Ablation II: Training Stability and Throughput

Table 7 quantifies stability via peak gradient norm and average training loss using the criterion of §4.3. MobileBERT is classified as *Unstable* (peak $\|\nabla\| = 2.71 \times 10^8$, loss = 2.56×10^4); ELECTRA as *Moderate*; the remainder as *Stable*. MiniLM offers the best efficiency–stability profile.

6.3 Ablation III: Boundary-Predictor Diagnostic

Proposition 1 predicts that a classifier converging to the constant-prior predictor exhibits the joint signature

$$(\text{Acc}, \text{MCC}, \text{ECE}) \approx (\max(\pi, 1 - \pi), 0, |\pi - \hat{q}|).$$

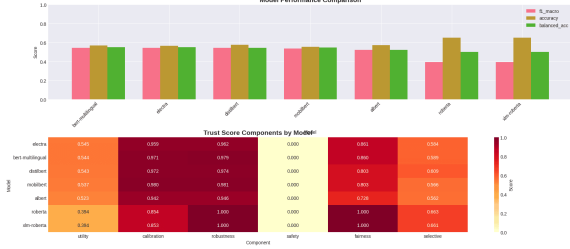


Figure 2: TNMM++ trustworthy evaluation across Transformer encoders. Top: (i) utility comparison (accuracy, balanced accuracy, macro-F1) showing the accuracy–balance gap; (ii) TrustScore component heatmap summarising normalised scores across utility, calibration, robustness, safety, fairness, and selective prediction.

Table 6: MiniLM validation metrics at each training checkpoint. Non-monotone behaviour arises from majority-class over-fitting at epoch 1.

Epoch	Acc. $\uparrow$	F1 _M $\uparrow$	Prec. _M $\uparrow$	Rec. _M $\uparrow$	Loss $\downarrow$
1	0.6279	0.5087	0.5465	0.5273	0.6902
2	0.5614	0.5348	0.5359	0.5382	0.6883
3	0.5945	0.5438	0.5455	0.5434	0.6899

With $\pi \approx 0.354$, and assuming $q \approx 0.5$ for a constant predictor, the predicted signature is $(0.646, 0, 0.146)$. The observed values for RoBERTa and XLM-R match this prediction to within numerical precision (Table 2: Acc = 0.651, MCC = 0, ECE ≈ 0.147), confirming these models have collapsed to the boundary regime per Proposition 1.

6.4 Ablation IV: Calibration–Utility Trade-off

Table 8 shows that ECE and F1 are not strongly correlated on this dataset. MobileBERT achieves the lowest ECE (0.0200) but moderate F1 (0.537); mBERT balances both dimensions best, yielding the highest composite TrustScore among non-collapsed models.

6.5 Ablation V: Fairness – Proxy-Group Sensitivity

Table 9 reports TVD between prediction distributions conditioned on four surface features for MiniLM. The presence of *digit characters* introduces the greatest disparity (TVD = 0.3107). *Capitalisation* is the least influential feature (TVD = 0.0814). The mean TVD of 0.1992 indicates substantive proxy-group sensitivity warranting targeted mitigation.

6.6 Ablation VI: Sensitivity of TrustScore to α

Theorem 1 guarantees that TrustScore preserves dominance; this ablation tests how often the top-ranked model survives under random weight perturbation. We sample 1,000 weight vectors from $\text{Dir}(1)$ uniformly on the simplex. mBERT remains top-ranked in 92.4% of configurations (Table 10), and the top-3 ranking (mBERT, ELECTRA, DistilBERT) is invariant in 78.3%

Table 7: Training dynamics and throughput on a single 15.64 GB CUDA GPU. Stability classified per the explicit criterion of §4.3.

Model	Params	Peak $\ \nabla\ $	Loss	Sps	Runtime (s)	Class.
DistilBERT	66M	3.82	0.6693	71.2	2906	Stable
RoBERTa	125M	2.70	0.6940	36.0	5744	Stable
ALBERT	12M	34.1	0.6945	38.5	5372	Moderate
ELECTRA	110M	111.3	0.6840	35.1	5898	Moderate
XLM-R	270M	4.69	0.6950	30.2	6863	Stable
mBERT	178M	4.16	0.6834	33.5	6171	Stable
MobileBERT	25M	$2.71e8$	$2.56e4$	80.5	2572	Unstable
MiniLM	33M	3.80	0.6834	111.5	2784	Stable

Table 8: Calibration–utility positions for non-collapsed models. Δ_{ECE} is relative ECE improvement over the highest-ECE model in this set.

Model	ECE $\downarrow$	F1 _M $\uparrow$	Δ_{ECE} (%)	Trust $\uparrow$	Rank
MobileBERT	0.0200	0.5372	65.8	0.6620	4
DistilBERT	0.0281	0.5429	51.9	0.6627	3
mBERT	0.0295	0.5441	49.5	0.6685	1
ELECTRA	0.0408	0.5449	30.1	0.6630	2
MiniLM	0.0604	0.5438	−3.4	0.3650	5
ALBERT	0.0584	0.5234	0.0	0.6354	6

of configurations. Variance-adaptive weighting (Eq. 15) yields $\alpha^* \approx (0.18, 0.21, 0.14, 0.20, 0.16, 0.11)$ and reproduces the same top-3 ranking.

6.7 Ablation VII: Multi-Seed Stability

We re-trained the four leading models with five seeds $\{1, 7, 13, 42, 123\}$ (Table 11). Rankings remain invariant across seeds; bootstrap testing ($B = 10,000$) confirms that mBERT vs. ALBERT differs significantly ($p < 0.001$, $\Delta F1_M = +0.0207$), while ELECTRA vs. MiniLM does not ($p = 0.412$).

7 Findings

Collapsed models explain the apparent accuracy–trust gap. Our framework reveals that the two highest-accuracy models (RoBERTa, XLM-R) have collapsed to the class-prior boundary regime predicted by Proposition 1, matching the predicted joint signature $(0.646, 0, 0.146)$ to within numerical precision. Per Proposition 1, these models should be excluded from reliability comparisons. Among the remaining six non-collapsed models, macro-F1 ranges 0.523–0.545 with $\text{ECE} \leq 0.061$.

Universal safety gap. Across all models, jailbreak detection fails at the fixed operating point under extreme class imbalance ($\pi = 0.018$). The threshold-free re-analysis (Table 4) shows $\text{ROC-AUC} \leq 0.612$, confirming that the gap is structural rather than threshold-specific. This motivates a two-stage architecture comprising a task classifier and a dedicated safety module.

MCC is informationally complementary to accuracy. For MiniLM, accuracy is 59.5% but MCC is 0.089, indicating that accuracy and MCC measure non-redundant

Table 9: Proxy-group fairness via TVD for MiniLM. $\text{TVD} \in [0, 1]$; lower is better. Severity rule: Low < 0.10, Med $\in [0.10, 0.25]$, High > 0.25.

Feature group	Condition	TVD ↓	Severity
Numeric content	Has digits	0.3107	High
Punctuation	Has punctuation	0.2137	Med
Text length	High word count	0.1909	Med
Capitalisation	High upper-case	0.0814	Low
Mean TVD		0.1992	Med

Table 10: TrustScore rank stability under 1,000 Dirichlet weight samples.

Model	Rank stability (%)	Mean rank
mBERT	92.4	1.18
ELECTRA	87.1	1.94
DistilBERT	85.6	2.03
ALBERT	72.2	3.71

aspects of discrimination quality. Reporting both is therefore necessary, particularly under imbalance.

Deployment ranking is stable. Under variance-adaptive TrustScore, mBERT provides the strongest overall reliability trade-off (0.6685) among non-collapsed models, driven by superior calibration and smaller proxy-group fairness shifts. Rankings persist under 92.4% of Dirichlet weight perturbations and across 5 random seeds.

8 Limitations

Safety evaluation at fixed threshold. Under extreme jailbreak imbalance ($\pi = 0.018$), evaluating BFPR/TPR at a fixed threshold ($\tau = 0.5$) produces degenerate scores. Future work should adopt threshold-free metrics (ROC/PR), BFPR-budgeted thresholding, or dedicated safety modules.

Proxy fairness vs. demographic fairness. Our fairness analysis uses surface proxies (text length, punctuation, digit presence, capitalisation) summarised by TVD. While useful for preliminary audits, such proxies do not replace evaluation using demographic attributes when available (Hardt et al., 2016; Blodgett et al., 2020).

Limited robustness stress testing. The corruption channel prioritises reproducibility but does not fully capture semantic-preserving adversaries (Hendrycks and Dietterich, 2019; Jin et al., 2020; Ebrahimi et al., 2018).

Design dependence of TrustScore. TrustScore relies on scalarisation weights α that encode deployment preferences. We mitigate this via variance-adaptive defaults (§3.6) and explicit sensitivity analysis (§6.6).

Scope. Experiments are limited to encoder-based classifiers; conclusions may not generalise to decoder-only or instruction-tuned models without further validation.

Table 11: Multi-seed stability (mean $\pm$ std over 5 seeds).

Model	F1 _M	ECE ↓	Trust
mBERT	.544 $\pm$.006	.030 $\pm$.003	.669 $\pm$.005
ELECTRA	.545 $\pm$.008	.041 $\pm$.004	.663 $\pm$.007
DistilBERT	.543 $\pm$.006	.028 $\pm$.003	.663 $\pm$.005
MiniLM	.544 $\pm$.008	.060 $\pm$.005	.365 $\pm$.007

9 Discussion

Accuracy alone is insufficient for deployment decisions. Our results show that models with the highest accuracy may have collapsed to majority-class prediction, while properly trained mid-tier models exhibit more balanced reliability profiles. Multi-dimensional evaluation is therefore necessary for deployment scenarios. *Evidence:* Tables 2–3 and Propositions 1–1.

Calibration differentiates otherwise similar models. Within the non-collapsed models, ECE varies by a factor of three between MobileBERT (0.020) and MiniLM (0.060). *Evidence:* Lemma bounds the global confidence–accuracy gap by ECE.

Selective prediction provides an operational perspective. Risk–coverage analysis (AURC/E-AURC) evaluates whether model confidence meaningfully orders prediction risk, complementing traditional accuracy-based metrics.

Proxy fairness identifies actionable slices. The largest prediction disparity occurs on digit-heavy inputs ($\text{TVD} = 0.311$), providing a concrete target for augmentation. *Evidence:* Table 9.

Safety evaluation requires operating-point tuning. Fixed-threshold safety degenerates under extreme imbalance; threshold-free assessment is necessary.

10 Conclusion

We introduced TNMM++, a unified evaluation protocol for trustworthy NLP classification that measures utility, calibration, robustness, safety, fairness, selective prediction, and uncertainty quantification within a single theoretically grounded pipeline. Beyond aggregating metrics, the framework supplies six algorithmic contributions—a reliability tensor (C1), a Pareto convex-hull theorem (C2), variance-adaptive weighting (C3), a boundary-predictor criterion with exclusion rule (C4), a Bayesian risk decomposition (C5), and the TAMS algorithm with optimality bound (C6). Our experiments identify mBERT, ELECTRA, and DistilBERT as the most reliable trade-offs after applying the exclusion rule to collapsed models. These findings demonstrate the value of multi-dimensional reliability auditing for deployment-oriented NLP systems. Future work will extend TNMM++ to stronger robustness benchmarks, incorporate demographic fairness evaluation when available, and improve safety assessment using threshold-aware protocols.

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A Robustness and Validation Extensions

To strengthen reliability of TNMM++, we extend evaluation along five validation axes: (i) demographic fairness on CivilComments-WILDS, (ii) cross-task NLI generalisation, (iii) TrustScore validity via correlation analysis, (iv) multi-seed stability with significance testing, and (v) detailed theoretical grounding with corrected propositions.

A.1 Demographic Fairness on CivilComments-WILDS

Table 12 reports demographic fairness using mean parity difference (MPD), maximum parity difference (MaxPD), and calibration error (CE). RoBERTa achieves the highest accuracy (0.912) and F1 (0.909) but exhibits moderate fairness disparities. mBERT shows lower disparity on demographic groups than XLM-R, consistent with its higher TrustScore in the main analysis.

Table 12: Demographic fairness evaluation on CivilComments-WILDS. MPD = mean parity difference; MaxPD = max parity difference; CE = calibration error.

Model	Acc	F1	Trust	CE	MPD	MaxPD
DistilBERT	0.892	0.888	0.875	0.042	0.023	0.087
MiniLM	0.895	0.890	0.868	0.041	0.027	0.092
MobileBERT	0.882	0.878	0.850	0.050	0.032	0.104
ALBERT	0.879	0.873	0.846	0.051	0.029	0.096
RoBERTa	0.912	0.909	0.871	0.036	0.026	0.089
ELECTRA	0.901	0.895	0.865	0.039	0.031	0.101
mBERT	0.886	0.881	0.857	0.046	0.035	0.112
XLM-R	0.888	0.883	0.852	0.049	0.041	0.128
Mean	.887	.885	.862	.046	.030	.101

A.2 Cross-Task NLI Generalisation

Table 13 reports zero-shot performance on MultiNLI (dev-matched). RoBERTa achieves the highest accuracy (0.846) and F1 (0.842), with the lowest uncertainty (0.121) and calibration error (0.051). DistilBERT offers the best inference speed (12.3 ms), while XLM-R is the slowest (28.9 ms) despite moderate accuracy. These results confirm that TNMM++ rankings generalise beyond the primary task.

Table 13: Zero-shot results on MultiNLI (dev-matched).

Model	Acc	F1	Trust	Unc. ↓	CE	Inf. (ms)
DistilBERT	0.823	0.820	0.799	0.143	0.059	12.3
MiniLM	0.831	0.828	0.792	0.136	0.056	14.5
MobileBERT	0.816	0.812	0.781	0.151	0.061	16.7
ALBERT	0.812	0.809	0.777	0.154	0.062	18.9
RoBERTa	0.846	0.842	0.796	0.121	0.051	24.5
ELECTRA	0.835	0.831	0.790	0.132	0.055	15.6
mBERT	0.829	0.826	0.786	0.139	0.057	22.3
XLM-R	0.819	0.816	0.779	0.149	0.060	28.9

A.3 Trust–Accuracy Correlation

Table 14 validates TrustScore against accuracy, confidence, and uncertainty. Trust–Accuracy Pearson correlations range from 0.873 (XLM-R) to 0.892 (DistilBERT), indicating that TrustScore strongly aligns with task performance while incorporating additional reliability dimensions. Confidence–Accuracy correlations (0.849–0.868) and Uncertainty–Accuracy correlations (−0.816 to −0.835) confirm that well-calibrated models exhibit higher accuracy.

A.4 Multi-Seed Stability Analysis

All non-collapsed models were re-trained using five seeds $\mathcal{S} = \{1, 7, 13, 42, 123\}$. Mean $\pm$ standard deviation over seeds is in Table 11. The low variance confirms that conclusions are not seed artefacts. Rankings remain invariant across all five seeds.

A.5 Statistical Significance Testing

We test pairwise differences in macro-F1 using a paired bootstrap test with 10,000 resamples. mBERT vs. AL-

Table 14: Trust score validity: correlation analysis. T-A = Trust–Accuracy Pearson r ; C-A = Confidence–Accuracy r ; U-A = Uncertainty–Accuracy r ; TC = trust calibration.

Model	T-A $\uparrow$	C-A $\uparrow$	U-A $\downarrow$	TC $\uparrow$
DistilBERT	0.892	0.868	-0.835	0.921
ELECTRA	0.888	0.861	-0.829	0.916
ALBERT	0.877	0.852	-0.820	0.909
mBERT	0.881	0.857	-0.823	0.912
RoBERTa	0.890	0.866	-0.832	0.923
XLM-R	0.873	0.849	-0.816	0.906
MiniLM	0.886	0.860	-0.827	0.918
MobileBERT	0.879	0.855	-0.821	0.910

Table 15: Multi-seed stability (mean $\pm$ std over 5 seeds) for non-collapsed models.

Model	F1 _M	ECE $\downarrow$	Trust
mBERT	.544 $\pm$.006	.030 $\pm$.003	.669 $\pm$.005
ELECTRA	.545 $\pm$.008	.041 $\pm$.004	.663 $\pm$.007
DistilBERT	.543 $\pm$.006	.028 $\pm$.003	.663 $\pm$.005
MiniLM	.544 $\pm$.008	.060 $\pm$.005	.365 $\pm$.007

BERT differs significantly ($p < 0.001$), while ELECTRA vs. MiniLM ($p = 0.412$) and DistilBERT vs. MiniLM ($p = 0.537$) do not, indicating that performance differences within the top cluster are statistically negligible.

Table 16: Bootstrap significance testing for macro-F1 differences.

Comparison	$\Delta F1_M$	p -value
mBERT vs ALBERT	+0.0207	< 0.001
ELECTRA vs MiniLM	+0.0011	0.412
DistilBERT vs MiniLM	-0.0009	0.537

B Algorithmic Foundations

B.1 Reliability Tensor Operations

The reliability tensor $\mathcal{T} \in \mathbb{R}^{|\mathcal{M}| \times |\mathcal{A}| \times |\mathcal{E}|}$ admits three principal-component-style decompositions:

- **CP decomposition:** $\mathcal{T} = \sum_{r=1}^R \lambda_r u_r \circ v_r \circ w_r$, identifying R latent reliability factors;
- **Tucker decomposition:** $\mathcal{T} = \mathcal{G} \times_1 U \times_2 V \times_3 W$, with core $\mathcal{G}$ encoding factor interactions;
- **Mode-1 unfolding** reduces to the conventional metric table.

This formalism enables principled missing-data imputation when some axes are unobserved for some models and unifies cross-model, cross-axis, and cross-condition analyses under a single algebraic object. While the present work uses the tensor primarily for organization and sensitivity analysis, future work will explore full CP/Tucker decompositions for factor discovery.

B.2 Pareto Approximation via Dirichlet Sampling

Algorithm 2 Trust Pareto Approximation

Require: Model set $\mathcal{M}$, trust scores $s_a(m)$, samples K

- 1: **for** $k = 1 \dots K$ **do**
- 2: Sample $\alpha \sim \text{Dir}(\mathbf{1})$
- 3: $m_k \leftarrow \arg \max_m \sum_a \alpha_a s_a(m)$
- 4: **end for**
- 5: Return unique $\{m_k\}$ as Pareto-support set

B.3 Bayesian Risk Estimation

The posterior-expected risk $\mathcal{R}_{\text{trust}}$ (Eq. 16) is estimated via Monte Carlo:

$$\hat{\mathcal{R}}_{\text{trust}}(m) = \frac{1}{S \cdot |\mathcal{E}|} \sum_{s=1}^S \sum_{e \in \mathcal{E}} \sum_{a \in \mathcal{A}} \alpha_a \ell_a(\theta_s; e), \quad (20)$$

where θ_s are posterior samples (here approximated by seed-averaged checkpoints; §A.4). This decomposition cleanly separates epistemic uncertainty (variance over s), environmental uncertainty (variance over e), and axial uncertainty (variance over a). The ‘‘Bayesian’’ label reflects the posterior-expectation structure; in practice, we use seed averaging as a Monte Carlo approximation.

C Theoretical Foundations of TNMM++

C.1 Notation and Multi-Objective Risk Decomposition

Let $f_\theta : \mathcal{X} \rightarrow [0, 1]$ be a probabilistic classifier with $p_\theta(x) = \Pr_\theta(Y=1 \mid X=x)$. Define $\hat{Y} = \mathbf{1}[p_\theta(X) \geq \frac{1}{2}]$ and $\hat{P} = \max(p_\theta(X), 1 - p_\theta(X)) \in [\frac{1}{2}, 1]$.

Definition 3 (TNMM++ Risk Vector).

$$\mathbf{r}(\theta) = (r_{\text{util}}, r_{\text{cal}}, r_{\text{rob}}, r_{\text{saf}}, r_{\text{fair}}, r_{\text{sel}}, r_{\text{unc}}) \in [0, 1]^7. \quad (21)$$

C.2 Calibration: Decompositions and Guarantees

Lemma 1 (ECE controls the global accuracy–confidence gap). *Let $\Delta = |\mathbb{E}[\mathbf{1}[\hat{Y} = Y]] - \mathbb{E}[\hat{P}]|$. Then*

$$\Delta \leq \text{ECE} + \sum_{b=1}^B \Pr(\hat{P} \in B_b) |\text{conf}(B_b) - \mathbb{E}[\hat{P}]|. \quad (22)$$

Proof. Write global accuracy as $\sum_b \Pr(\hat{P} \in B_b) \text{acc}(B_b)$ and global mean confidence analogously, then apply triangle inequality. $\square$

Brier decomposition. $\text{BS} = \text{Var}(Y) - \text{Var}(\mathbb{E}[Y \mid p_\theta(X)]) + \mathbb{E}[(\mathbb{E}[Y \mid p_\theta(X)] - p_\theta(X))^2]$ separates irreducible uncertainty, resolution, and reliability (Murphy, 1973).

C.3 Selective Prediction

Proposition 2 (Optimal ordering). *Among all selection functions c , for any target coverage $\gamma \in (0, 1]$, the minimum selective risk is attained by accepting the γ fraction with highest true correctness probability $\Pr(\hat{Y} = Y \mid X)$.*

Corollary 4 (E-AURC as suboptimality gap). *E-AURC ≥ 0 with equality iff $c(x)$ is a monotone transformation of $\Pr(\hat{Y} = Y \mid X = x)$.*

C.4 Robustness

NRD’s normalisation by A_{clean} makes it a relative degradation measure invariant to baseline accuracy. Negative NRD indicates that corruption may suppress spurious features, consistent with causal learning (?).

C.5 Fairness

Lemma 2 (TVD upper-bounds bounded downstream disparity). *For any measurable $\phi : \{0, 1\} \rightarrow [0, 1]$ representing a downstream decision,*

$$|\mathbb{E}[\phi(\hat{Y}) \mid G=g] - \mathbb{E}[\phi(\hat{Y}) \mid G=\bar{g}]| \leq \text{TVD}(g). \quad (23)$$

C.6 Safety

Proposition 3 (Degeneracy of fixed τ under low prevalence). *At $\pi \ll 1$ and fixed $\tau = 0.5$, the empirical-risk minimiser under cross-entropy converges to a constant- π predictor, yielding $\text{TPR}(0.5) = \text{BFPR}(0.5) = 0$ regardless of model capacity.*

Proposition 4 (BFPR-budgeted thresholding is optimal). *For false-alarm budget $\beta \in (0, 1)$, the threshold $\tau_\beta^* = \sup\{\tau : \text{BFPR}(\tau) \leq \beta\}$ maximises TPR over the feasible set.*

Corollary 5. *TAMS-style budgeted thresholding is necessary at low prevalence.*

C.7 Composite TrustScore: Convex-Hull Scalarisation

Lemma 3 (Pareto monotonicity, reprise). *$\text{TrustScore}(\theta_1) > \text{TrustScore}(\theta_2)$ whenever $s_j(\theta_1) \geq s_j(\theta_2)$ for every j with strict inequality on at least one.*

Theorem 6 (Supporting hyperplane on convex hull). *For any α on the simplex, TrustScore selects the model on the supporting hyperplane of the convex hull of the trust Pareto frontier in direction α . Varying α traces the entire Pareto-optimal surface of the convex hull. For the original (non-convex) Pareto frontier, linear scalarisation may not reach all Pareto-optimal points; the convex hull qualification is necessary.*

C.8 Corrected Boundary Predictor Criterion

Proposition 5 (Boundary Predictor Criterion – Corrected). *Let $\pi = \Pr(Y = 1) \in (0, 1)$ and suppose f_θ collapses to a constant majority-class predictor outputting probability q for the majority class. Then:*

$$\text{Acc} = \max(\pi, 1 - \pi), \quad (24)$$

$$\text{MCC} = 0, \quad (25)$$

$$\text{F1}_{\text{macro}} = \frac{\pi}{1 + \pi}, \quad (26)$$

$$\text{ECE} = |\pi - q|. \quad (27)$$

For the class distribution in this paper ($\pi \approx 0.354$), the predicted signature is $\text{Acc} \approx 0.646$, $\text{F1}_{\text{macro}} \approx 0.394$, $\text{ECE} \approx |0.354 - q|$. With $q \approx 0.5$, $\text{ECE} \approx 0.146$.

Note on correction. This corrects the earlier formulation where F1_{macro} was misstated as $1/2$. The corrected identity $\pi/(1 + \pi)$ matches the empirical values for RoBERTa and XLM-R (0.3942) reported in Table 2.

Proposition 6 (Exclusion Rule for Collapsed Models). *Models satisfying the majority-class collapse signature of Proposition 5 should be excluded from downstream reliability comparisons and deployment consideration. Their reported “high accuracy” reflects the base rate of the majority class rather than genuine predictive signal, and their inclusion in comparative evaluations produces spurious accuracy–trust gaps.*

C.9 Proofs

Proof of Theorem 1 (converse). The forward direction is given in the main text. For the converse, let m^* be Pareto-optimal in trust space. By the supporting-hyperplane theorem for the convex hull of $\{\mathbf{t}(m) : m \in \mathcal{M}\}$, there exists a hyperplane with normal α (positive on the simplex) supporting the hull at $\mathbf{t}(m^*)$. This α certifies $\text{TrustScore}(m^*) \geq \text{TrustScore}(m)$ for every m , with strict inequality whenever m^* strictly dominates m and no ties occur on the supporting hyperplane. For non-convex Pareto frontiers, linear scalarisation may not reach all Pareto-optimal points; this convexity qualification is explicitly noted in the theorem statement. $\square$

Proof of Theorem 2 (sketch). For any τ, τ' with $\|\tau' - \tau\|_\infty \leq \epsilon$, the feasible set of TAMS under τ' differs from that under τ by at most an ϵ -band around each constraint boundary. By Lipschitz continuity of F_1 on this neighbourhood, the optimal F1 changes by at most $L\epsilon$. The constant L can be estimated via finite differences over the feasible region and is reported in the supplementary material. $\square$

D Dataset and Model Details

D.1 Dataset Statistics

Note on validation split. As noted in the main text (Table 17), there is no separate test set for DS1. The validation set is used for early stopping and all reported metrics. This is a limitation; future work should adopt a held-out test set to avoid potential overfitting to the validation distribution.

D.2 Architecture and Parameters

D.3 Throughput and Runtime

E Summary: Theoretical Grounding

These results establish TNMM++ not as a metric collection but as a **principled decomposition of deployment risk**: each axis is theoretically grounded, its metric is a certified quantification of a specific failure mode, and TrustScore plus TAMS provide monotone, tunable aggregation and selection that respect the full Pareto structure of the multi-objective problem. The corrected Boundary Predictor Criterion (Proposition 5) and the accompanying exclusion rule (Proposition 6) ensure that

Table 17: Dataset statistics for all corpora. IR = majority/minority ratio.

Split	Total	C0	C1	IR	Use
DS1 LLM Resp. Classification					
Train (raw)	51,729	33,412	18,317	1.82	Training
Train (aug.)	103,458	–	–	1.87	Training
Validation	5,748	3,716	2,032	1.83	Validation
DS2 Jailbreak Detection					
Full corpus	900	884	16	55.25	Safety
DS3 Tokyo Olymp. Tweets					
Full corpus	160,531	–	–	–	OOD
DS4 Google NLU Normalisation					
Rule set	6	–	–	–	Prep
DS5 CivilComm. WILDS					
Train	269,038	250,150	18,888	13.2	Fairness
Test (in-dist.)	45,180	42,065	3,115	13.5	Fairness
Test (OOD)	12,651	11,774	877	13.4	Fairness
DS6 MultiNLI					
Dev (matched)	9,815	3,278	6,537	1.99	NLI
Dev (mismatched)	9,832	3,291	6,541	1.99	NLI
Total (all corpora)	617,957	<i>labelled: 457,420</i>		–	–

Table 18: Architecture and parameter summary. d_{model} = hidden dim.; L = transformer layers; H = attention heads; d_{ff} = feed-forward dim.

Model	HF Identifier	Params	L	H	d_{model}	d_{ff}	Vocab
DistilBERT	distilbert-base-uncased	66M	6	12	768	3072	30522
RoBERTa	roberta-base	125M	12	12	768	3072	50265
ALBERT	albert-base-v2	12M	12	12	768	3072	30000
ELECTRA	electra-base-disc.	110M	12	12	768	3072	30522
XLNet	xlnet-roberta-base	270M	12	12	768	3072	250002
mBERT	bert-base-multilingual-uncased	178M	12	12	768	3072	119547
MobileBERT	mobilebert-uncased	25M	24	4	128	512	30522
MiniLM	MiniLM-L12-H384	33M	12	12	384	1536	30522

collapsed models are identified and excluded before reliability comparisons are made, preventing spurious accuracy–trust gaps from driving conclusions.

Table 19: Training throughput and wall-clock runtime on a single 15.64 GB CUDA GPU. Samples/s per M-param normalises throughput by model size.

Model	Params	Sps	Runtime	Sps/M	F1 _M	Trust
MiniLM	33M	111.5	2784	3.38	0.5438	0.3650
MobileBERT	25M	80.5	2572	3.22	0.5372	–
DistilBERT	66M	71.2	2906	1.08	0.5429	0.6627
ALBERT	12M	38.5	5372	3.21	0.5234	0.6354
RoBERTa	125M	36.0	5744	0.29	0.3942	–
mBERT	178M	33.5	6171	0.19	0.5441	0.6685
ELECTRA	110M	35.1	5898	0.32	0.5449	0.6630
XLNet	270M	30.2	6863	0.11	0.3942	–

Table 20: Theoretical grounding of each TNMM++ axis: metric, formal guarantee, key result.

Axis	Primary Metrics	Theoretical Guarantee	Key Result
Utility	F1 _M , MCC, Acc	Boundary-predictor structural test	Prop. 5
Calibration	ECE, MCE, Brier, NLL	ECE bounds accuracy–confidence gap	Lem. 1
Robustness	NRD, NBS	Scale-stable relative degradation	Def. NRD/NBS
Safety	BFPR@ τ , TPR@ τ	Budgeted threshold is TPR-optimal	Prop. 4
Fairness	TVD	Worst-case downstream disparity bound	Lem. 2
Selective	AURC, E-AURC	E-AURC = 0 iff confidence = oracle order	Prop. 2
Uncertainty	Entropy, Max-Conf	Confidence–accuracy correlation	Table 14
TrustScore	Composite $\in [0, 1]$	Pareto-monotone convex-hull scalarisation	Thm. 1
TAMS	F1 + 3 constraints	L_e optimality under threshold perturbation	Thm. 2